

# Secure and Trustworthy Data Management Mechanism for Dance-Consumer Electronics in AIoT

Hua Wei, Junfeng Miao<sup>✉</sup>, Jianhui Lv<sup>✉</sup>, Chien-Ming Chen<sup>✉</sup>, *Senior Member, IEEE*,  
Saru Kumari<sup>✉</sup>, *Senior Member, IEEE*, and Mohammed Amoon<sup>✉</sup>

**Abstract**—The integration of Artificial Intelligence (AI) with the Internet of Things (IoT) marks a considerable advancement in data collection and utilization across various sectors. In the realm of complex scenarios, dance-consumer electronics (DCE) employ this technology to provide detailed and trustworthy documentation and three-dimensional digital capture of dancers' movements. However, the handling of sensitive physiological data poses significant security and trustworthiness risks in such consumer electronics, including potential breaches that could be exploited for location tracking and behavior prediction. To address the challenges of encrypted traffic detection and the need for trustworthiness in IoT, this paper introduces the AI-Enhanced Anonymous Traffic Filtering Framework, which features terminal anonymization. This framework employs the Inception-SelfAttBiGRU model to filter malicious traffic and includes a credibility-based filtering mechanism prior to this module to reduce network load and enhance system efficiency for those consumer electronics. The evaluation concentrated on two main areas: assessing the access credibility-based filtering mechanism's ability to identify and block malicious DCE endpoints and evaluating the effectiveness of the encrypted traffic classification technique. Using simulated environments and public datasets, this comprehensive assessment demonstrated that the dual-layer protection mechanism effectively detects and blocks malicious traffic, thereby enhancing system security and maintaining network integrity.

**Index Terms**—AIoT, consumer electronics, security and trustworthiness, encrypted traffic filtering.

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Hua Wei is with Department of Music, Xinxiang University, Xinxiang 453003, China (e-mail: weihua36@xxu.edu.cn).

Junfeng Miao is with the College of Computer and Information Engineering, Henan Normal University, Xinxiang 453007, China (e-mail: miaojunfengwu@gmail.com).

Jianhui Lv is with the Department of Network, Peng Cheng Laboratory, Shenzhen 518057, China (e-mail: lvjh@pcl.ac.cn).

Chien-Ming Chen is with the School of Artificial Intelligence, Nanjing University of Information Science and Technology, Nanjing 210044, China (e-mail: chienmingchen@ieee.org).

Saru Kumari is with the Department of Mathematics, Chaudhary Charan Singh University, Meerut 250004, India (e-mail: saryusirohi@gmail.com).

Mohammed Amoon is with the Department of Computer Science, Community College, King Saud University, Riyadh 11437, Saudi Arabia (e-mail: mamoon@ksu.edu.sa).

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## I. INTRODUCTION

THE INTEGRATION of Artificial Intelligence (AI) with the Internet of Things (IoT), commonly referred to as AIoT, represents a significant advancement in how data is collected and utilized across various sectors [1]. AIoT systems harness a wide array of sensors to collect real-time data—including environmental, operational, business, and monitoring data—which is then intelligently processed and analyzed at the device, edge, or cloud levels using data mining and machine learning techniques [2]. This technology has increasingly permeated national infrastructure and public services, impacting fields ranging from science to the humanities, including the performing arts such as dance. In dance, AIoT technologies facilitate comprehensive and precise documentation and three-dimensional digital capture of dancers' movements, enabling detailed analysis of their techniques and experiences to derive patterns and insights [3]. The convergence of AIoT with dance not only enhances the way performances are executed and experienced but also opens new avenues for exploring interactive narrative structures in digital dance arts [4]. This paper defines such applications within typical dance scenarios as dance-consumer electronics (DCE), which include devices like smart dance shoes, motion monitoring wristbands, fitness trackers, motion capture systems, and interactive dance technology systems.

The DCE system utilizes IoT sensing devices to collect user data, and uploads it to local or cloud servers for analysis and processing [5]. However, the collection of sensitive physiological data such as heart rate, movement speed, and body temperature presents significant security and trustworthy challenges [6]. These data are intimately tied to an individual's health and personal details, making unauthorized access a serious threat that could lead to trustworthiness breaches and potential misuse for purposes such as location tracking and behavior prediction. To mitigate these risks, encryption of data traffic has become a common practice, ensuring that even if attackers intercept the data, they cannot read or analyze the sensitive information. This not only shields the data from prying eyes but also significantly reduces the potential for data misuse. Moreover, many of these devices lack robust security measures. Outdated software can exacerbate security flaws, leaving systems susceptible to exploitation [7]. The consequences of such vulnerabilities can be severe; compromised

devices might be hijacked to dispatch malicious traffic, leading to Distributed Denial of Service (DDoS) attacks that disrupt network communication and system operations [8]. This could severely impact the performance monitoring systems crucial in dance environments.

Classifying and detecting malicious encrypted traffic effectively mitigates security challenges in Dance-consumer electronics [9]. There are two main technological approaches for detecting encrypted traffic: cryptographic-based malicious traffic identification and artificial intelligence (AI)-based malicious traffic identification [10]. Cryptographic identification utilizes the distinct statistical features manifested by ciphertexts produced under different encryption schemes to detect malicious traffic. This method is particularly effective for traffic with known encryption characteristics but less so for traffic with unknown features. On the other hand, AI-based identification involves extracting features from target encrypted traffic, using these features to train classification models, and optimizing the models to achieve high detection accuracy. This approach can effectively detect malicious traffic regardless of the encryption algorithm used, providing a robust solution to the problem of unknown encrypted traffic.

This paper introduces an AI-enhanced anonymous traffic filtering framework tailored to address the challenges of detecting encrypted malicious traffic and preserving the trustworthiness of traffic characteristics. At the core of this framework is the Inception-SelfAttBiGRU model, which is specifically chosen for its robust ability to handle the complexities of encrypted traffic. This model transforms raw traffic data into traffic images, effectively utilizing both spatial and temporal feature extraction techniques. The Inception module within this model is particularly well-suited for processing traffic images at varying scales, allowing for the extraction of intricate patterns with minimal computational overhead. This capability is crucial for real-time processing in environments where security is a priority. Complementing this, the Self-Attentive BiGRU component is adept at capturing temporal dependencies and nuances in traffic flow. Together, these components process the traffic images to extract deep learning features, resulting in protocol-independent feature vectors that are critical for detailed encrypted traffic analysis. To enhance security and operational efficiency, the framework also integrates access anonymization management and a credibility filtering mechanism. These components work by anonymizing endpoint identities and rigorously evaluating the credibility of each endpoint. Traffic from sources deemed untrustworthy is effectively blocked, thereby reducing the network load and increasing the overall effectiveness of the system. This strategic blocking of traffic not only fortifies the security of the network but also ensures that only credible, verified traffic is processed, aligning with the framework's goal to maintain the integrity and trustworthiness of network data. The primary contributions of this paper are threefold:

- 1) **Introduction of the AI-Enhanced Anonymous Traffic Filtering Framework:** This framework leverages the Inception-SelfAttBiGRU model to transform raw traffic into traffic images for encrypted traffic analysis. It

provides a novel approach to detect encrypted malicious traffic using deep learning techniques.

- 2) **Integration of Access Anonymization and Credibility Filtering:** The framework integrates access anonymization management and a filtering mechanism based on access credibility, which obscures endpoint identities and evaluates the consistency of identity verification and behavior with the access context. This dual-layer protection mechanism enhances the system's security and efficiency.
- 3) **Comprehensive Evaluation and Validation:** The effectiveness of the proposed framework is evaluated using both simulated environments and publicly available datasets. The experimental results demonstrate that the dual-layer protection mechanism can effectively detect and block malicious traffic, thus improving system security and ensuring the integrity of the network.

This paper is structured as follows: Section II reviews related work. Section III outlines our AI-enhanced anonymous traffic filtering framework, including its three main modules: access anonymization, credibility filtering, and cryptographic traffic filtering. Section IV assesses the framework's effectiveness using simulations and public datasets. Section V concludes our paper.

## II. RELATED WORK

### A. Access Control for IoT Devices

Access control for IoT devices is increasingly crucial as the Internet of Things expands and smart device usage proliferates, accentuating the need for robust trustworthiness protection. Access control mechanisms safeguard against unauthorized access and operations by delineating permissions across varying user levels, thus securing sensitive data effectively. Reference [11] highlights a smart contract-based role access control model that supports autonomous, decentralized application functioning and responsive resource management, enhancing access control flexibility and security. However, improvements in the granularity and dynamism of access permissions are suggested. Ateniese et al. [12] developed a secure distributed storage solution using Proxy Re-Encryption (PRE), allowing only data owners with a master private key to access encrypted files. This mechanism's ability to transform ciphertext into forms readable by designated users has led to its application across various contexts, as noted in studies [13], [14], [15]. Further security enhancements have been realized, with Libert and Vergnaud [16] developing a scheme resistant to Replayable Chosen Ciphertext Attacks (RCCA). In the context of IoT, Hong and Sun [17] devised an attribute-based PRE scheme that incorporates a key isolation mechanism, ensuring the forward security of user private keys and re-encryption keys, which facilitates secure and finely granulated data sharing. In another significant contribution, Kim and Lee [18] proposed a method for efficiently managing and sharing data on lightweight devices using proxy re-encryption, effectively reducing the computational load and enhancing the functional capabilities of networks with limited resources.

### B. Encrypted Traffic Detection and Classification

Encryption, a technology that enhances the data's information content while reducing predictability, affects the statistical characteristics of the data without altering its length, thereby increasing its randomness [19]. Academically, this increase in randomness is exploited to distinguish between encrypted and non-encrypted traffic effectively. Niu et al. [20] proposed the Heuristic Statistical Testing (HST) method, utilizing various randomness tests to measure traffic randomness, with the C4.5 decision tree classifier for detecting encrypted traffic.

In parallel, machine learning and deep learning have revolutionized fields such as computer vision and natural language processing and have shown substantial benefits in encrypted traffic classification. Wang et al. [21] leveraged deep learning's end-to-end learning capabilities for automated feature selection using a 1D-CNN. Song et al. [22] innovated further by integrating Text-CNN for classification, addressing the hierarchical nature of encrypted traffic and introducing a novel loss function and category weight distribution to smooth the classification process. Additionally, the ICLSTM model by Lu et al. [23], which merges Inception CNN with LSTM, exemplify advanced integrations for enhanced performance. Xie et al. [24] incorporated the Self-attention mechanism with CNN to refine feature selection, demonstrating superior efficacy compared to conventional models. Lotfollahi et al. [25] and Izadi et al. [26] respectively utilized stacked autoencoders and convolutional neural networks to achieve high recall rates in network traffic classification tasks, highlighting the effectiveness of deep learning structures in handling complex classification scenarios.

Furthermore, encrypted malicious traffic detection has also progressed with methods like those by Yu et al. [27] using Multi AutoEncoders for feature extraction and detection. De Lucia and Cotton [28] and Yang et al. [29] also contributed significantly, using SVM, CNN, and Bayesian neural networks to handle the nuanced challenges of encrypted malicious traffic detection, thereby advancing the field towards more reliable and effective solutions. Zeng et al. [30] employs a hybrid deep learning model combining LSTM and CNN for enhanced detection capabilities.

Our framework integrates deep learning and access control to address encrypted traffic detection and secure access management. Unlike previous works, our AI-enhanced anonymous traffic filtering framework incorporates the Inception-SelfAttBiGRU model to transform raw traffic into traffic images for encrypted traffic analysis, combined with terminal anonymization and access credibility filtering to enhance security and efficiency. This approach leverages deep learning strengths and robust access control for IoT environments.

## III. AI-ENHANCED ANONYMOUS TRAFFIC FILTERING FRAMEWORK

### A. Overview of Our Method

To tackle the identified security and trustworthiness issues, this study introduces an AI-enhanced anonymous traffic

filtering framework, aimed at strengthening the trustworthiness of data in Dance-Consumer Electronics in AIoT environments. This method involves anonymizing endpoint identities to obscure traffic characteristics from the detection systems, crucial for identifying malicious activities within encrypted communications. The framework employs a dual-layered strategy that integrates access anonymization, evaluation of device credibility, and advanced deep-learning techniques to filter cryptographic traffic effectively.

The proposed framework includes three modules in total, namely the access anonymization module, credibility filtering module, and cryptographic traffic filtering module. The first module, access anonymization, assigns anonymized identities to each DCE endpoint, creating a secure and trustworthy environment for traffic detection. This approach prevents adversaries from inferring the identities of DCE endpoints based on their traffic patterns, thereby safeguarding traffic trustworthiness. The second module, credibility filtering, involves assessing the credibility of source endpoints. This process blocks traffic from sources deemed untrustworthy, effectively filtering out malicious traffic at its origin. Credibility is evaluated based on how well the endpoint's identity verification and its behavior conform to the access context. For traffic stemming from endpoints of moderate credibility, challenging to assess, redirection to the third module for filtering the uncertain traffic. The third module utilizes the Inception-SelfAttBiGRU model. It transforms raw traffic into images, extracts spatial and temporal features using Inception modules and BiGRU with Self-attention, respectively, and classifies these into protocol-independent vectors via a softmax classifier for detecting encrypted malicious traffic efficiently. The Inception module is ideal for this task due to its ability to process images at varying scales and extract intricate patterns without extensive computational costs, which is critical for real-time processing in security-sensitive environments. Meanwhile, the Self-Attentive BiGRU component enhances the model by providing the capability to capture temporal dependencies and nuances in traffic flow, which are often missed by conventional convolutional networks. The encrypted traffic detection service can be performed by one or more servers within the scenario that possess strong computing capabilities, thereby alleviating the computational burden on the DCE devices.

### B. Filtering Mechanism Based on Terminal Access Credibility

1) *Concealed Access for Terminals*: Concealed access for DCE endpoints facilitates the protection against malicious inference of their identities based on traffic characteristics, while preserving the integrity of these characteristics. The process of anonymizing endpoint access is bifurcated into two distinct phases: the generation of anonymous identities and their subsequent verification. This section details the creation of these concealed identities and their authentication process to ensure secure and anonymous access.

**Creation of Concealed Terminal Identities.** Given that servers often display semi-trustworthy and inquisitive

behaviors, demonstrating interest in decrypting and examining traffic flows and possibly circumventing certain safeguards to gain direct access to unprocessed data. Thus, to maintain a securely and trustworthily anonymous setting, the servers, within the identity creation and confirmation cycle, should be restricted to serving merely as conduits for message relay. Anonymous identity creation using proxy re-encryption consists of the following five steps.

*Step 1:* Each participant begins by generating a unique random number  $n$ , which serves as a temporary anonymous parameter. The participant then constructs an anonymity request in the form

$$R(T_{ID}) = E_{T_p}(n \| T_{ID} \| ST_s(n) \| t_R) \| r_{T-A},$$

where  $T_p$  denotes the participant's public key,  $T_{ID}$  the true identity,  $ST_s(n)$  the signature over  $n$ , and  $r_{T-A}$  the re-encryption key derived from the participant's private key  $T_s$  and the server's public key  $A_p$ ,  $t_R$  is the timestamp indicating when the request is made.

*Step 2:* Upon receiving the request, the server uses  $r_{T-A}$  to re-encrypt the data, obtaining  $R(T_{ID})_r$ , and forwards this to the anonymity server. The anonymity server decrypts the payload with its private key  $A_s$  and verifies whether  $n$  is listed in its database of anonymous parameters.

*Step 3:* If  $n$  is not listed, the anonymity server creates a new anonymous identity

$$P_{ID} = h(T_{ID} \| n),$$

and logs this identity alongside the real identity in a private blockchain accessible solely to the anonymity server. If  $n$  is found in the database, indicating a repeat, the anonymity server issues a reset command to the participant to select a different  $n$ , ensuring the uniqueness of each anonymous identity.

**Step 4:** Next, the anonymity server packages the anonymous identity  $P_{ID}$  along with its digital signature  $SA_s(P_{ID})$  and a timestamp  $t_s$  into a response, encrypts it using its public key  $A_p$ , and attaches the re-encryption key  $r_{A-T}$ , which is generated using the server's private key  $D_s$  and the anonymity server's public key  $A_p$ , before sending it back to the server. This package confirms the identity is generated by the anonymity server. The encrypted response is structured as follows:

$$S(P_{ID}) \equiv E_{A_p}(P_{ID} \| SA_s(P_{ID}) \| t_s) \| r_{A-T}$$

*Step 5:* The server then broadcasts the encrypted response  $S(P_{ID})_r$ , which has been re-encrypted using  $r_{A-T}$ . Each participant uses their private key  $T_s$  to decrypt the response and retrieve their unique anonymous identity  $P_{ID}$ .

The creation phase ensures that each terminal is assigned a unique anonymous identity, preventing unauthorized tracking or inference of terminal behavior. Once identities are created, the next critical step is to authenticate these identities to confirm their legitimacy and grant access to network resources.

**Authentication of Concealed Terminal Identities.** The process for authenticating the anonymous terminal is shown as follows.

*Step 1:* The access terminal initiates the process by sending an anonymous identity verification request

$$R_V(P_{ID}) = (P_{ID} \| ST_s(P_{ID}) \| t_{R_v}) \| r_{T-A}$$

to the server. Here,  $P_{ID}$  represents the terminal's anonymous identity,  $ST_s(P_{ID})$  is the signature over  $P_{ID}$  by the terminal,  $r_{T-A}$  is a re-encryption key generated using the terminal's private key  $T_s$  and the identity of the target server, and  $t_{R_v}$  is the timestamp for the authentication request.

*Step 2:* The server uses the re-encryption key  $r_{T-A}$  to calculate the re-encrypted ciphertext  $R_V(P_{ID})_r$  and forwards it to the anonymous server. The anonymous server decrypts  $R_V(P_{ID})_r$  using its private key  $A_s$  and searches for  $P_{ID}$  in the anonymous identity mapping table.

*Step 3:* If  $P_{ID}$  is found in the table, the verification result is set as  $V_{PID} = 1$ . The anonymous server then responds with

$$S_V(V_{PID}) \equiv E_{D_p}(V_{PID} \| SA_s(V_{PID}) \| t_{S_v})$$

where  $D_p$  is the public key of the server,  $SA_s(V_{PID})$  is the signature on the verification result, and  $t_{S_v}$  is the response time. If  $P_{ID}$  is not found,  $V_{PID} = 0$ , and the anonymous server sends a rejection response to the terminal.

*Step 4:* The server decrypts  $S_V(V_{PID})$  using its private key  $D_s$  to obtain the verification result  $V_{PID}$ . If  $V_{PID} = 1$ , the server issues an access token to the terminal, enabling it to join the current network with this token.

**2) Assessment of Trust for Terminal Access:** The evaluation of a terminal's access reliability involves an analysis of its access context and prior activities, along with how well the terminal's anonymous verification and environmental interactions are understood. This process aims to preclude malicious traffic from the outset. Terminals deemed highly reliable are considered to transmit credible traffic, which is allowed to pass. In contrast, terminals with low reliability are presumed to be sources of malicious traffic and are consequently blocked.

**Quantification of Access Trust.** To streamline the assessment of access trust, the terminal's trust score is normalized to the  $[0, 1]$  interval. The Beta probability density function stands out for its computational ease, adaptability, and efficiency, making it an apt model for trust calculation, represented as follows:

$$\text{Beta}(\alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1} \quad (1)$$

The parameter  $\theta$  quantifies the action probability, which is the likelihood of a device engaging in a specific behavior. It lies within the open interval  $(0,1)$ . The parameters  $\alpha$  and  $\beta$  are critical to shaping this probability distribution. They represent the number of successful and unsuccessful anonymous identity verifications, respectively. Both parameters must be positive.

Terminals are subjected to anonymous identity verification, which forms the basis for evaluating their initial credibility. If the anonymous identity validation succeeds for the  $i^{\text{th}}$  terminal user  $V'_{i1}$ , and not for  $V'_{i2}$ , then the foundational trust level of IIoT terminals is derived from a cumulative statistic of anonymous validations, expressed as:

$$T_{\text{BASE}_i} = \frac{V_{i1} + 1}{V_{i1} + V_{i2} + 2} \quad (2)$$



Anonymous identity verification serves as a fundamental factor in assessing the reliability of IoT terminal access. For the  $i$ -th terminal, let  $V_{i1}$  and  $V_{i2}$  represent the counts of successful and failed anonymous verifications, respectively.

This formula adjusts for extreme values by incorporating a pseudocount, enhancing stability in reliability assessments. However, this base reliability metric may exhibit delays in detecting compromised terminals, as it does not adapt quickly to sudden changes in device behavior, potentially keeping the trust index within a narrow fluctuation range for extended periods. In dynamic environments, such as those involving dance-consumer electronics (DCE) terminals, trust should evolve with user interactions and environmental changes, not remain static.

To accommodate the dynamic nature of IoT environments, we refine the evaluation method by introducing an access statistical count  $c(k)$ , which accumulates with each device interaction, reflecting the device's growing familiarity with its environment:

$$c(k) = c(k-1) + \frac{1}{\tau} \quad (3)$$

where  $\tau$  represents the access threshold, and  $k$  is the number of access attempts. This count is reset to zero once the threshold  $\tau$  is reached, ensuring responsiveness to recent access patterns.

Additionally, we define trust ( $f_p$ ) and suspicion ( $f_d$ ) coefficients to capture the nuances of terminal interactions over time. These coefficients are computed as follows:

$$f_p = \sum_{k=1}^{n-1} (e(k) \cdot c(k)) \quad (4)$$

$$f_d = \sum_{k=1}^{n-1} (d(k) \cdot c(k)) \quad (5)$$

where  $e(k)$  measures the similarity between the current and historical access environments, and  $d(k)$  measures their differences. These metrics are adjusted based on the degree of overlap with historical data, where  $m$  is the count of similar past accesses.

The comprehensive access credibility  $T_{COMP_i}$  of the terminal is then adjusted by the threat attenuation factor  $t_i$ , calculated as:

$$t_i = \frac{1}{\text{mean}(F_i)} \sqrt{\sum_{j=1}^n f_{pj}^2 + f_{dj}^2} \quad (6)$$

where  $F_i$  represents the historical set of trust and suspicion coefficients for the  $i$ -th terminal. This factor sensitively adjusts the base trust index based on the variance in access conditions, as reflected in the coefficients.

The final credibility index is expressed conditionally, accounting for the attenuation:

$$T_{COMP_i} = \begin{cases} \frac{V_{i1}}{V_{i1}+V_{i2}+2} \cdot e^{-t_i}, & \text{if } T_{BASE_i} > t_i \\ 0, & \text{otherwise} \end{cases} \quad (7)$$

This approach allows for a more dynamic adjustment of trust based on real-time data, making it more suitable for

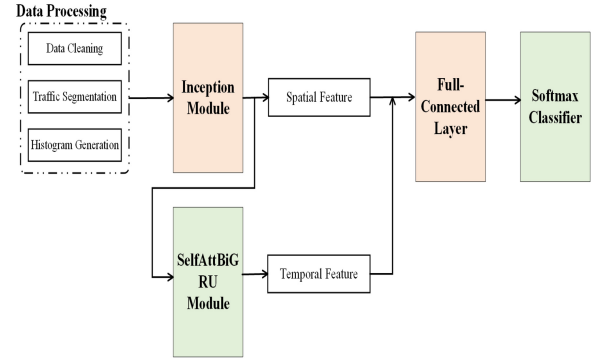


Fig. 1. System Architecture of Inception-SelfAttBiGRU Model.

environments where DCE devices interact frequently and under varying conditions.

Thus, the traffic filtering mechanism based on terminal access credibility can be summarized in three main steps:

*Step 1:* Initiate by verifying the terminal's anonymous identity. Extract the signature  $ST_s(P_{ID})$  from encrypted traffic and authenticate it through an anonymous server. If authentication fails, block the terminal's traffic and increment the failure count:  $V_{i2} = V_{i2} + 1$ . If successful, increment the success count:  $V_{i1} = V_{i1} + 1$ .

*Step 2:* Determine the terminal's comprehensive credibility  $T_{COMP_i}$ . Start by computing the base credibility  $T_{BASE_i}$  using the outcome of the identity verification. Then, pull historical access data from the record table  $Y_c$ , and evaluate how the current access context compares to past data. Calculate the trust coefficient  $f_p$  and suspicion coefficient  $f_d$  based on whether the current environment is new, the same, or partially overlapping with historical ones, assigning  $e(k)$  and  $d(k)$  accordingly. Calculate the threat attenuation factor  $t_i$  based on these coefficients to finalize  $T_{COMP_i}$ .

*Step 3:* Assess whether to allow the traffic through or forward it to the next filtering layer. If  $T_{COMP_i}$  falls below a set threshold  $\delta$ , the traffic is deemed trustworthy and is permitted to proceed. Otherwise, it is escalated for further scrutiny.

### C. Encrypted Traffic Filtering Mechanism Based on Deep Learning

Following the terminal access credibility assessment, which blocks potentially malicious terminals, this section introduces a complementary approach focusing on raw network traffic. The deep learning-based encrypted traffic filtering mechanism leverages advanced neural architectures to enhance security against both identity-based and content-based threats. This paper introduces the Inception-SelfAttBiGRU model, which transforms raw network traffic into traffic images using packet arrival time and packet size as axes. These images are then processed through a deep learning model for feature extraction. Spatial features are captured using Inception modules, while temporal features are extracted with BiGRU and Self-attention mechanisms. The resulting protocol-independent feature vectors are classified using a fully connected layer and a softmax classifier.

1) *Data Processing*: The process of the data processing employed converts traffic data into two-dimensional histograms, providing a visual representation of the data distribution and a more intuitive understanding of its characteristics. This paper positions time on the horizontal axis and frame size of traffic data on the vertical axis, with frame direction indicated by varying shades, which helps in observing traffic distribution over different time periods, differences in upstream and downstream frame distribution, and the relationship between traffic volume and time. This method is superior to single-dimensional traffic sequence analysis as it captures the density and directional differences in frame sizes, revealing the multidimensional complexity of the data. The two-dimensional histogram offers a comprehensive description of data distribution, detailing range, concentration, and density patterns, which enhances feature representation capability. This approach effectively captures the distribution characteristics and trends of frame sizes in link-layer encrypted traffic, laying a robust foundation for further identification and classification. The specific steps in the processing workflow are as follows:

**Data Cleaning.** The data cleaning step is critical for ensuring the accuracy and reliability of the subsequent analysis. Initially, this process involves removing corrupted or incomplete data frames from the dataset. Any frames that do not conform to expected traffic protocols or that exhibit anomalies in their metadata, such as timestamps or source/destination addresses, are discarded. This step also includes the normalization of frame sizes and timestamps to ensure uniformity across the dataset. Furthermore, outliers that significantly deviate from typical data patterns are identified and evaluated to determine whether they represent genuine traffic phenomena or data collection errors.

**Traffic Segmentation.** Data frames collected with the same source and destination addresses are grouped into the same data stream. An “observation window” is used to represent a segment of the stream for statistical analysis, extracting features from each window separately. The observation window, defined as  $\pi$ , maps the traffic time series as  $T_1, T_2, \dots, T_\pi, T_{\pi+1}, \dots, T_N$ . Without sliding windows, traffic data are divided into segments such as  $\{T_1, T_2, \dots, T_\pi\}$ ,  $\{T_{\pi+1}, T_{\pi+2}, \dots, T_{2\pi}\}$  and so forth. The use of sliding window techniques, where segments such as  $\{T_1, T_2, \dots, T_\pi\}$ ,  $\{T_2, T_3, \dots, T_{\pi+1}\}$  are analyzed, allows for a richer sampling of features while ensuring variability among samples. The chosen time window is 30 seconds, with a sliding step of 3 seconds.

**Histogram Generation.** Each time window is used to extract data frame sizes and their arrival times relative to the first frame, serving as the horizontal and vertical coordinates, respectively. Traffic originating from mobile devices to routers is defined as upstream and represented by positive values, while downstream traffic to mobile devices is represented by negative values. Normalization is applied to map frame sizes between  $-50$  and  $+50$ , and time windows between  $0$  and  $100$ , resulting in a  $100 \times 100$  two-dimensional histogram. Each cell represents the count of frames with a specific size during the corresponding time interval, with the sum of all

cells representing the total number of frames captured in that 30-second sample.

2) *Spatial Feature Extraction Using the Inception Module*: Convolutional Neural Networks (CNNs) are deep neural network models designed to process data with a grid-like topology. Traditional CNNs often increase network depth to extract complex features, which requires extensive parameters and computational resources. This approach leads to bulky models that are inefficient in practical applications. Moreover, they may fail to capture multi-level features within the input data, limiting model performance. The Inception module addresses these limitations by using convolutions at various scales to achieve nuanced feature extraction across different levels of abstraction.

In addition to the multi-scale convolutional architecture, this paper enhances the Inception module with Batch Normalization (BN), aimed at mitigating issues like vanishing gradients, exploding gradients, and speeding up network convergence. BN normalizes the input layers by adjusting the input distributions to a standard form, maintaining a stable distribution across the neural network’s layers which improves both the training efficiency and generalization of the model. Given a batch of input data  $x_1, x_2, \dots, x_m$  where  $m$  denotes the batch size, the normalization process involves calculating the mean  $\mu$  and variance  $\sigma^2$  for the inputs to a layer, normalizing these inputs, and then applying a scale and shift transformation. This transformation uses trainable parameters  $\gamma$  and  $\beta$  to adjust the data, as outlined in Equations (8)-(11), where  $\epsilon$  is a constant.

$$\mu = \frac{1}{m} \sum_{i=1}^m x_i \quad (8)$$

$$\sigma^2 = \frac{1}{m} \sum_{i=1}^m (x_i - \mu)^2 \quad (9)$$

$$\hat{x}_i = \frac{x_i - \mu}{\sqrt{\sigma^2 + \epsilon}} \quad (10)$$

$$y_i = \gamma \hat{x}_i + \beta \quad (11)$$

Following normalization, data is processed through the ReLU activation function, defined in Equation (12). This function applies a simple threshold to the input, enhancing computational speed in practical applications, especially for large-scale deep neural networks.

$$f(x) = \begin{cases} 0, & \text{if } x \leq 0 \\ x, & \text{if } x > 0 \end{cases} \quad (12)$$

3) *Temporal Feature Extraction Using the SelfAttBiGRU*: While the Inception network adeptly learns distributed feature representations across different scales in parallel, it does not adaptively adjust inter-channel weights and often suffers from significant discrepancies in the relevance of the feature maps generated by each branch. This can lead to improper feature selection.

To leverage both the temporal and spatial features of traffic images, Recurrent Neural Networks (RNNs) are employed for temporal feature extraction. However, traditional RNNs struggle with vanishing gradients and cannot retain long-term information effectively. The Gated Recurrent Unit (GRU), an

advanced version of the RNN, introduces gate mechanisms that control information flow between hidden states, thus alleviating the issues of gradient explosion and vanishing in RNNs. To fully harness temporal features from both preceding and succeeding contexts, Bidirectional Gated Recurrent Units (BiGRU) are utilized.

Furthermore, considering the varying importance of each packet in session traffic, a Self-attention mechanism is integrated on top of the BiGRU model to calculate weights for the outputs of the last hidden layer through weighted summation. This enhances the accuracy and efficiency of the model. The structure of the BiGRUSelf-attention model comprises four main layers, which are input layer, BiGRU layer, hidden layer and self-attention layer. Initially, features extracted by the Inception module are input as  $e_n$ . These features are fed into both forward and backward GRU cells of the BiGRU layer to extract temporal features. Each input thus generates a forward hidden output  $\vec{h}_n$  and a backward hidden output  $\overleftarrow{h}_n$ . For self-attention layer, the forward and backward hidden output are added firstly.

$$h_n = \vec{h}_n \oplus \overleftarrow{h}_n \quad (13)$$

This output forms the input vector  $H = [h_1, h_2, h_3, \dots, h_n]$  for the Self-attention layer, where  $Q$ ,  $V$ , and  $K$  values are computed:

$$Q = H \times W^Q \quad (14)$$

$$V = H \times W^V \quad (15)$$

$$K = H \times W^K \quad (16)$$

Attention weights are then calculated using the following equation:

$$a = \text{softmax}\left(\frac{Q \cdot K^T}{\sqrt{D_k}}\right) \quad (17)$$

Here,  $D_k$  represents the dimension of either the query vector  $Q$  or the key vector  $K$ . The final output is obtained as follows:

$$h_n = \sum_{j=1}^N v_j \cdot a_j \quad (18)$$

The combination of BiGRU and Self-attention mechanisms successfully extracts temporal features from the traffic images.

4) *Encrypted Traffic Classification*: This paper proposes an encrypted traffic classification model that integrates Inception networks with BiGRU Self-attention mechanisms. The model begins with preprocessing two-dimensional histogram matrices of the traffic data which are then inputted into the model for training. Initially, two sets of convolution and pooling layers extract primary features from the input images, distributing global information across multiple local feature matrices. This setup enables the extraction of learned image features directly from input images. Subsequent layers consist of several Inception modules that process these feature maps. Different-sized convolutional kernels operate in parallel within these modules to capture both detailed and global information from the two-dimensional histograms, enhancing the perception of spatial features related to data frame size, direction, and

arrival time. Additionally, SelfAttBiGRU module can aid in the accurate recognition and selection of key attributes. An adaptive average pooling operation then reduces the spatial dimensions of the feature maps to  $1 \times 1$ , creating a global feature representation. Finally, the Flatten layer transforms the multi-dimensional feature tensor into a one-dimensional vector. This vector feeds into a fully connected neural network that classifies the encrypted traffic. To enhance the model's generalization ability and robustness, a dropout layer is incorporated, which randomly deactivates some neurons to reduce the model's dependency on specific neurons, effectively preventing overfitting. Together, these components form a deep learning network model capable of effectively distinguishing between malicious and benign traffic samples. For encrypted traffic that is challenging to classify based on access credibility assessments alone, this model can classify encrypted traffic by combining the Inception module and the SelfAttBiGRU module, enabling effective security and trustworthy for DCE terminals.

## IV. EXPERIMENTAL EVALUATIONS

### A. Experimental Settings

To evaluate the effectiveness of the methodologies described herein, experimental analyses are conducted on a trustworthy traffic filtering framework to assess its capability in detecting malicious encrypted traffic. The evaluation comprised two primary components: initially, the access credibility of DCE endpoints is examined to ascertain the efficacy of the proposed method in identifying malicious DCE endpoints. Subsequently, the performance of the encrypted traffic classification method is appraised. The experiments utilized a hardware setup featuring an Intel Core i7-10700 processor, 16 GB of RAM, and the Windows 10 operating system. The deep learning model employed is based on Pytorch 1.7.0 and accelerated using an NVIDIA RTX3090 GPU.

A simulated environment involving a multi-person dance scenario, such as a dance troupe, is used to evaluate the terminal detection scheme based on credibility. The scenario involved 400 DCE terminals, aimed at thoroughly documenting and accurately capturing the dancers' movements and physiological states. To assess system resilience against malicious attacks, 35% of the DCE devices are configured as dishonest terminals, simulating control by attackers. It is hypothesized that these devices would engage in malicious traffic from the 40th of 120 communication rounds. A credibility threshold of  $\delta=0.4$  is established.

For the performance evaluation of the deep learning-based encrypted traffic classification method, we conducted experiments using the ISCX-VPN-nonVPN dataset from the Canadian Institute for Cybersecurity. This dataset includes seven types of session traffic and their encrypted counterparts using protocol encapsulation. The sessions encompass a diverse range of applications: Email-VPN (emails sent via SMTP and received through POP3 or IMAP), Chat-VPN (various instant messaging apps), Streaming-VPN (streaming media services), File Transfer-VPN (file sharing under multiple protocols), VOIP-VPN (Voice Over IP calls),

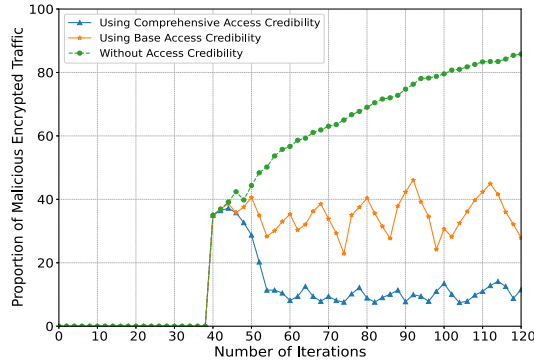


Fig. 2. Suppression of Malicious Traffic Using Different Access Credibility based-Filtering Mechanisms.

and P2P-VPN (peer-to-peer file sharing via protocols like Bittorrent). The initial category of Browsing-VPN, characterized by traffic generated through Web browsing, is excluded from the evaluations due to the high variability and non-distinctive patterns of browsing behavior, which often overlap with other categories, leading to unclear classification boundaries compared to other categories. The study is hence limited to the six remaining types of encrypted session traffic. For the experimental validation, 30% of the data is randomly selected as the test set, with the remaining used for training.

### B. Evaluations on Filtering Mechanism Based on Terminal Access Credibility

To observe the impact of employing different access credibility-based filtering mechanisms on suppressing malicious traffic from compromised terminals, we measured the variation of the proportion of malicious encrypted traffic across iterations, as depicted in Figure 2. The use of basic access credibility led to some degree of suppression but is accompanied by considerable fluctuations and an upward trend, indicating a modest effect. In contrast, comprehensive access credibility consistently and significantly reduced the proportion of malicious traffic, maintaining a low level throughout the experiment. Without the implementation of any access credibility, the proportion of malicious traffic incrementally rose. With the number of iterations, this proportion exceeded 80% and continued to rise gradually, highlighting the destructive impact of compromised devices on the network. In summary, by integrating the attenuation factor  $t_i$  into the basic access credibility, comprehensive credibility promptly responds to the compromised behavior of IIoT terminals, thereby effectively curbing the surge in malicious traffic.

### C. Evaluations on Encrypted Traffic Filtering Mechanism Based on Deep Learning

In the evaluation of binary classification models, four principal metrics are utilized: Accuracy, Precision, Recall, and F1 Score. The definitions are as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \quad (19)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (20)$$

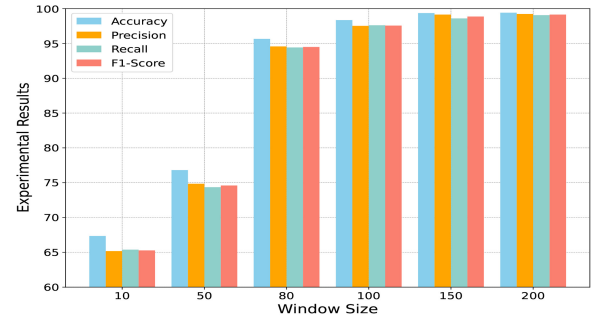


Fig. 3. Experimental Results of Our Method under Different Window Sizes.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (21)$$

$$F1score = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (22)$$

These metrics are computed using the quantities of true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN). For multi-class classification, the direct application of Precision, Recall, and the F1 Score is not straightforward, as they are inherently designed for binary scenarios. Hence, we adopt their macro-averaged counterparts: macro-Precision, macro-Recall, and macro-F1 Score. Each of these is the arithmetic mean of the respective metric calculated for each class independently.

In this study, we employed a sliding window method to segment the original traffic dataset. To explore the impact of varying window sizes on model performance, we set window size  $\pi$  to different values. The experimental results are shown in Figure 3. The results indicate that as the window size increases, there are significant improvements in model accuracy, precision, recall, and F1-score. Notably, the performance enhancements are particularly marked when the window size is increased from 10 to 150. For larger window sizes, all metrics reach or exceed 99%, demonstrating that the model can process and predict data with extremely high precision under these settings. Ultimately, this study selected a window size of 150. This choice is based on experimental data, as this window size maintains high levels of accuracy and other metrics while avoiding the additional computational overhead that larger windows might entail.

To evaluate the performance enhancement effects of different improvement modules within the model proposed in this paper, we designed the following variant models. Except for the specified modifications, all other parameters in these models are kept consistent. The comparative analysis involved four distinct network architectures: Standalone CNN, CNN+SelfAtBiGRU, Standalone Inception, and Inception+GRU. In the Standalone CNN model, the architecture employs only conventional convolutional layers without the advanced feature extraction capabilities of the Inception module or the dynamic temporal processing of the SelfAtBiGRU mechanism. This model serves as a baseline to evaluate the effectiveness of adding more complex structures. The CNN+SelfAtBiGRU architecture enhances the basic CNN model by integrating a Self-Attention mechanism with a Bidirectional GRU. This



TABLE I  
COMPARATIVE PERFORMANCE ANALYSIS OF DIFFERENT  
NETWORK ARCHITECTURES

| Network Model           | Accuracy | Precision | Recall | F1-Score |
|-------------------------|----------|-----------|--------|----------|
| Standalone CNN          | 90.23    | 89.79     | 89.42  | 89.60    |
| CNN+SelfAttBiGRU        | 95.68    | 95.21     | 94.92  | 95.06    |
| Standalone Inception    | 97.64    | 97.70     | 97.35  | 97.52    |
| Inception+GRU           | 98.45    | 98.90     | 98.12  | 98.51    |
| Inception+ SelfAttBiGRU | 99.36    | 99.14     | 98.59  | 98.86    |

addition aims to improve the model's ability to capture temporal relationships and dependencies in the data, expected to enhance performance in scenarios where sequence and timing are crucial. Standalone Inception employs the Inception module alone, leveraging its ability to perform multi-level feature extraction through parallel convolutional operations. This model tests the hypothesis that spatial feature diversity within the traffic images can significantly improve classification accuracy. Finally, the Inception+GRU model combines the Inception module with a unidirectional GRU. This configuration is designed to benefit from the Inception module's detailed feature extraction capabilities while also utilizing GRU to capture temporal features, aiming for a balanced approach to handling both spatial and temporal aspects of the traffic data. This structured approach ensures that each model variant focuses on testing the impact of specific components, thereby isolating the effects of each module on the overall performance of the network. Comparative experimental results are presented in Table I.

The experimental results indicate clear differences in performance across the various network models tested. The Standalone CNN, serving as a baseline, achieved scores with an accuracy and F1-score just under 90%. Introduction of the SelfAttBiGRU mechanism in the CNN+SelfAttBiGRU model markedly improved all metrics, indicating the efficacy of combining convolutional neural networks with bidirectional GRU and self-attention mechanisms for this application. Further enhancements are observed with the Standalone Inception model, which outperformed the Standalone CNN, underscoring the advantages of the Inception architecture in handling more complex features independently. The Inception+GRU model integrated these features with unidirectional GRU to further boost performance, achieving nearly 98.5% in both accuracy and F1-score, demonstrating significant gains particularly in precision. The most pronounced improvements are seen in the proposed Inception+SelfAttBiGRU model, which combined the strengths of Inception modules, bidirectional GRU, and self-attention. This configuration not only achieved the highest scores across all metrics—close to or exceeding 99%. But it also confirmed that the synergistic integration of these modules can substantially improve performance in complex model architectures. This model's superior precision and F1-score, in particular, highlight its robustness in both identifying relevant features and accurately classifying data points. These findings suggest that the addition of advanced architectural elements like the Inception module and SelfAttBiGRU significantly enhances the model's ability

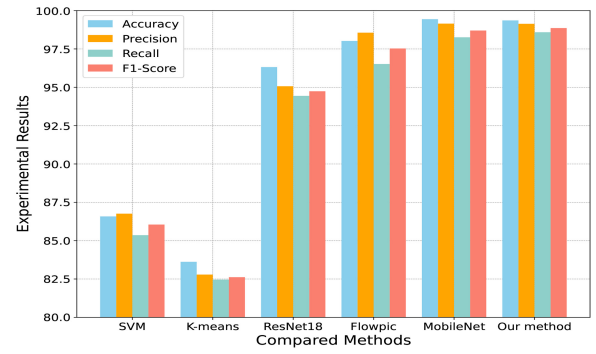


Fig. 4. Compared Experimental Results with Different Methods.

to process and interpret complex datasets more effectively than simpler CNN structures.

In a comparative analysis against established methods, our approach, which employs the Inception+SelfAttBiGRU architecture, demonstrated robust performance across all evaluated metrics. Support Vector Machines (SVM) are supervised learning models with associated learning algorithms that analyze data for classification and regression analysis. They are widely used due to their effectiveness in high-dimensional spaces and their versatility in modeling diverse sources of data. K-means clustering, a method of vector quantization, originally from signal processing, that is popular for cluster analysis in data mining. K-means clustering partitions  $n$  observations into  $k$  clusters in which each observation belongs to the cluster with the nearest mean, serving as a prototype of the cluster. Traditional methods such as Support Vector Machines (SVM) and K-means clustering yielded lower results, with SVM achieving an accuracy and F1-score of approximately 86%, and K-means slightly lower at around 83% for both metrics. ResNet18, a variant of the ResNet architecture that uses 18 layers, is known for its ability to enable training of deeper networks by utilizing skip connections or shortcuts to jump over some layers. Typical applications include image recognition tasks. The ResNet18 also underperformed relative to our method, achieving an accuracy close to 96% and a precision of approximately 95%. While ResNet18 represents a significant advancement over simpler architectures, it still did not match the more advanced systems in this evaluation. Flowpic is a method that visualizes network flow data as images, enabling the use of image processing techniques for network traffic analysis. It transforms flow data into a grid-like structure where different traffic characteristics are represented in various dimensions of the grid. More contemporary architectures like Flowpic and MobileNet showed closer performance to our method. Flowpic achieved commendable results, especially in precision at nearly 98.6%, but with a slightly lower recall which impacted its overall F1-score. MobileNet is a class of efficient models for mobile and edge devices. It is based on a streamlined architecture that uses depth-wise separable convolutions to build lightweight deep neural networks. MobileNet slightly outperformed our method in accuracy and precision but had a lower recall and corresponding F1-score. Our method, employing the Inception+SelfAttBiGRU

architecture, excelled in this evaluation, achieving an accuracy close to 99.4%, with high scores in precision and F1-score as well. This indicates a well-balanced capability in accurately identifying and classifying features within encrypted traffic. The integration of Inception modules with Self-Attention and BiGRU mechanisms proved particularly effective, providing the high precision and balanced recall necessary for the complex task of encrypted traffic classification.

## V. CONCLUSION

To address the security and trustworthiness challenges faced by Consumer Electronics (DCE), this paper presents an AI-enhanced anonymous traffic filtering framework that anonymizes DCE endpoints and employs encrypted traffic communication to prevent malicious entities from inferring terminal identities based on traffic characteristics. The framework integrates a credibility-based filtering mechanism and the Inception-SelfAttBiGRU model. To assess the effectiveness of the proposed secure and trustworthy framework in detecting malicious encrypted traffic, a comprehensive evaluation is conducted using both simulated environments and the publicly available ISCX-VPN-nonVPN dataset. The evaluation focused on two primary aspects: the performance of the access credibility-based filtering mechanism at identifying and blocking malicious DCE endpoints, and the efficacy of the encrypted traffic classification technique. Compared to basic access credibility and scenarios without any access control implementation, this comprehensive credibility approach rapidly addresses compromised IoT terminal behavior, effectively reducing the surge in malicious traffic. Besides, results on ISCX-VPN-nonVPN datasets show that the integration of Inception modules with Self-Attention and BiGRU mechanisms proved particularly effective, providing the high precision and balanced recall required for the complex task of encrypted traffic classification.

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